

AYMANE AIT DADS

Machine Learning Engineer Intern - Image Generation | Computer Vision · Generative AI · Deep Learning

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision and completion-only loss. Achieved 1st place in the EURECOM class leaderboard for AI-generated image detection (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) on 250K images spanning 25+ generative model types, then submitting to the NTIRE 2026 @ CVPR international benchmark. Bringing proven expertise in PyTorch, transformer fine-tuning, ensemble methods, and controlled ablation studies directly applicable to Genies' AI avatar and image generation pipeline. Eager to contribute to the visual and embodiment layer of AI personas by advancing generative image modeling, training efficiency, and production-grade ML engineering at Genies.

CORE COMPETENCIES

Image Generation | Generative AI | Computer Vision | Transformer Fine-Tuning | Deep Learning | Model Evaluation | Ablation Studies | ML Pipeline Engineering

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and deployed a **Random Forest + K-Means ML pipeline** on 100K+ fiber-optic network records to classify network quality across thousands of nodes with automated feature engineering.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production decisions.
- Reduced manual analysis time by **~60%** through end-to-end automated model evaluation and feature engineering, accelerating reporting cycles for network maintenance teams.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging data science results with business and operations teams; stack: Python, Scikit-learn, Pandas, Power BI.

PROJECTS

AI-Generated Image Detection - NTIRE 2026 *EURECOM ImSecu Course + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class · 0.791 AUC @ CVPR*

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head on 250K images spanning 25+ generative model types; submitted to NTIRE 2026 @ CVPR CodaBench international benchmark.
- Discovered that **1-epoch fine-tuning (0.793 AUC) outperforms 4-epoch training (0.767 AUC)** for OOD generalization; applied 10-view TTA (flips, crops, blur, resizes) and 3-model ensemble weighted by validation AUC using FP16 mixed precision and dual learning rates.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Cloud Computing, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | OpenCLIP | Unsloth | TRL | PEFT | Scikit-learn

Computer Vision & Generative AI: CLIP ViT-L/14 | CNN | DenseNet | image classification | LoRA SFT | BF16 mixed precision | TTA ensemble

Data Science & MLOps: Python | Pandas | NumPy | feature engineering | ablation studies | Git | Power BI